Camera Capture Using OV7675 Camera

Objective:

To capture an image frame using the OV7675 camera module connected to the Arduino Nano 33 BLE and transmit the pixel data through the Serial Monitor.

(Note: Install Arduino_OV767X Library)

Procedure:

1. Connected the OV7675 camera module to the Arduino Nano 33 BLE.
2. Installed the Arduino_OV767X library.
3. Opened the CameraCapture example:
4. **File → Examples → Arduino_OV767X → CameraCapture**
5. Uploaded the code to the board.
6. Opened the Serial Monitor.
7. Sent the character **'c'** through the Serial Monitor.
8. The camera captured one frame and transmitted the image data as a hexadecimal string.

Working Principle:

The OV7675 camera captures an image and converts it into pixel values.

The image is captured in

- Resolution: **176 × 144 pixels (QCIF)**
- Format: **RGB565**
- Pixel Size: **16 bits per pixel**

Each pixel contains color information represented as hexadecimal values. These pixel values are transmitted through the Serial Monitor.

Observations:

- Camera successfully captured image data.
- Pixel values were transmitted as hexadecimal numbers.
- The captured data represents the image frame.